

Problem

Great Migration

Problem ID: greatmigration

The swans are preparing for their great migration and need to maximise their energy reserves for the long flight. Because they need to stay light enough to fly, a swan can only carry a maximum weight of food in their digestive tract. The swans have a limited selection of aquatic plants available to eat. Each plant has a specific weight and yields a specific amount of migratory energy. A swan can choose to eat a plant or leave it, but they cannot eat a fraction of a plant.

Input

The first line of the input consists of two integers representing the number of plants (N) and the total stomach capacity (W), bounded by $(1 \leq N \leq 1000)$ and $(1 \leq W \leq 5000)$.

Each of the next N lines consists of two integers per line representing the weight (w_i) and energy value (v_i) of each plant, bounded by $(1 \leq w_i \leq W)$ and $(1 \leq v_i \leq 10^4)$.

Output

Output a single integer representing the maximum total energy the swan can accumulate without exceeding its weight capacity.

Sample Input 1

3 10
4 5
5 6
6 8

Sample Output 1

13

Sample Input 2

3 10
6 9
5 6
5 6

Sample Output 2

12

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